



Solving the Lorenz ODE System using Optimal ANN Architectures

Can a **plain feedforward ANN**, with no physics in the loss, learn a coupled nonlinear ODE system, and **which architecture does it best?**



Ikramul Hasan MoradMd. Abu BakarSamiur Rahman OmlanFariha IslamMd. Touhidul Islam

SUPERVISOR
Dr. Muhammad Nomani Kabir
Professor, Dept. of CSE, UIU

COURSE TEACHER
Dr. Md. Motaharul Islam
Professor, Dept. of CSE, UIU

TEAM NAME
Team_ParaDox
Final Year Design Project I

Introduction & Motivation

Coupled nonlinear ODEs like the **Lorenz-1960** system rarely admit closed-form solutions. Classical Runge–Kutta solvers are accurate but **mesh-dependent**, re-integrating step by step for every query. A trained network can instead learn the continuous map $t \rightarrow (\mathbf{x}, \mathbf{y}, \mathbf{z})$ in one shot.

Central question: can a **plain feedforward ANN**, trained only on numerical data with no physics in the loss, approximate the Lorenz-1960 solution, and **which architecture does it best?**

Objectives

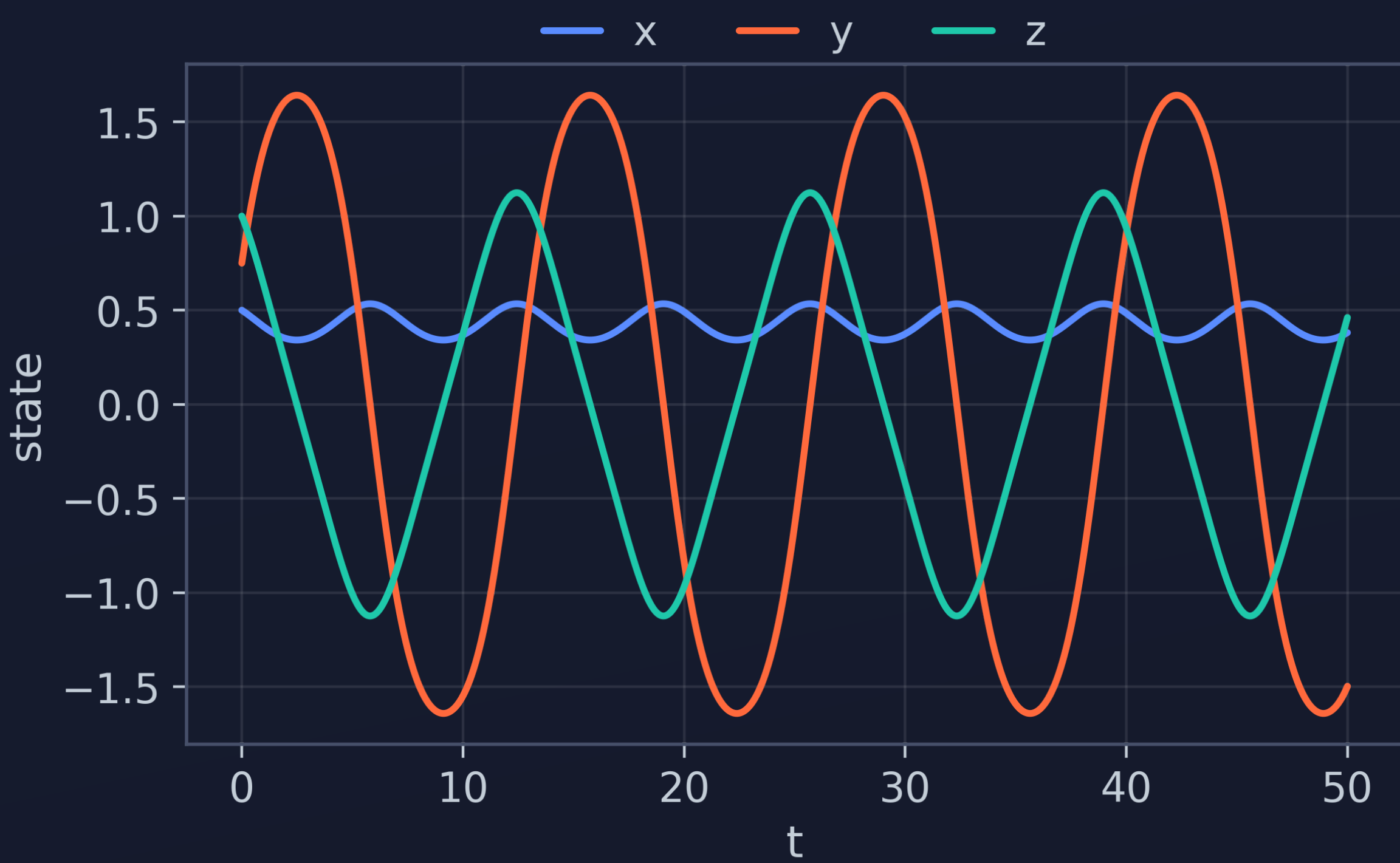
- Build a trusted reference with **RK4** and **SciPy**; cross-check one solver against the other.
- Score accuracy vs cost: MSE, MAE, max error, training / inference time.
- Systematically vary **depth**, **width**, and **activation**, with no physics aids in the loss.
- Benchmark the best plain ANN against **PINN** / **DeepONet** results from the literature.

Background: The System

$$\begin{aligned}\dot{x} &= -0.10 \, y \, z \\ \dot{y} &= +1.60 \, x \, z \\ \dot{z} &= -0.75 \, x \, y\end{aligned}$$

$k = 2, \ell = 1 \quad \cdot \quad x_0=0.5, y_0=0.75, z_0=1 \quad \cdot \quad t \in [0, 1]$

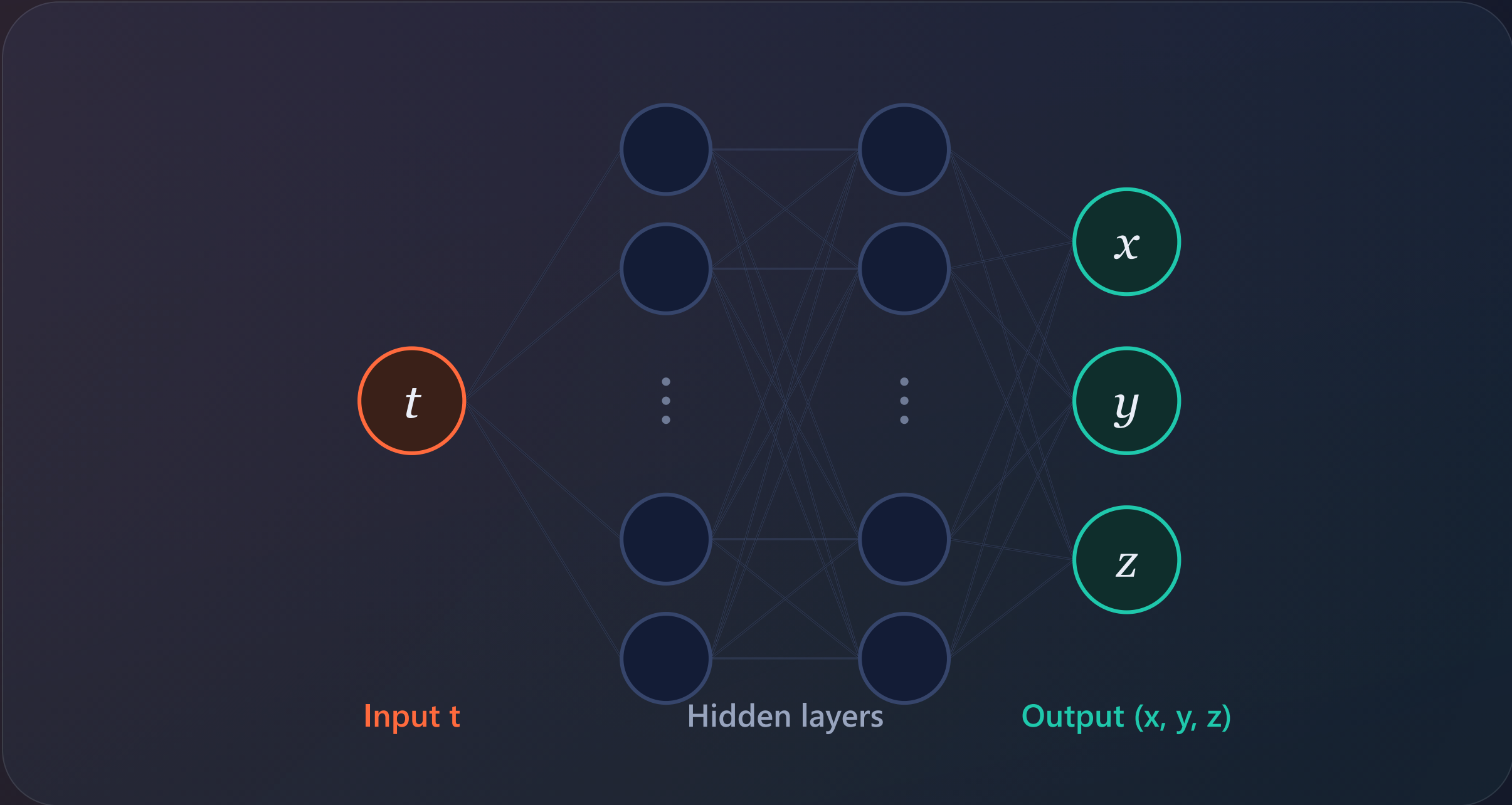
Three coupled nonlinear ODEs from an early meteorological model. Compact to write down, but hard to integrate accurately.



Bounded, quasi-periodic orbits over $t \in [0, 50]$.

The Network & the Search Space

60 architectures



One input $t \rightarrow$ stacked hidden layers $\rightarrow$ output (x, y, z) .

Depth
HIDDEN LAYERS: 1, 2, 3, 4

Width
NEURONS / LAYER: 20, 50, 100

Activation
NONLINEARITY: tanh, ReLU, sigmoid, GELU, Swish

One sweep over **depth × width × activation**, trained on numerical data with no physics in the loss. The three strongest designs advance to the optimizer study.

Research Gap & Questions

Gap. No prior work runs a systematic plain-ANN architecture study on Lorenz-1960; existing studies fix the architecture, use only simple ODEs, or rely on physics-informed losses.

What's missing

- No architecture study for Lorenz-1960
- No PINN vs plain-ANN comparison on it
- Depth **and** width rarely varied together

Research questions

- Can a plain ANN learn the solution from numerical data alone?
- Which depth × width × activation wins the accuracy–cost trade-off?
- How close does it come to physics-informed baselines?

Methodology: Two-Phase Study

69 controlled runs

Phase 1 · Architecture search

Sweep depth × width × activation with the optimizer held fixed (Adam).

= 60 runs

Phase 2 · Optimizer comparison

3 best Phase-1 architectures × {Adam, L-BFGS, SGD + momentum}, all else constant.

= 9 runs

Generate
RK4 + SciPy



Preprocess
80/20 split



Train
ANN grid



Evaluate
5 metrics

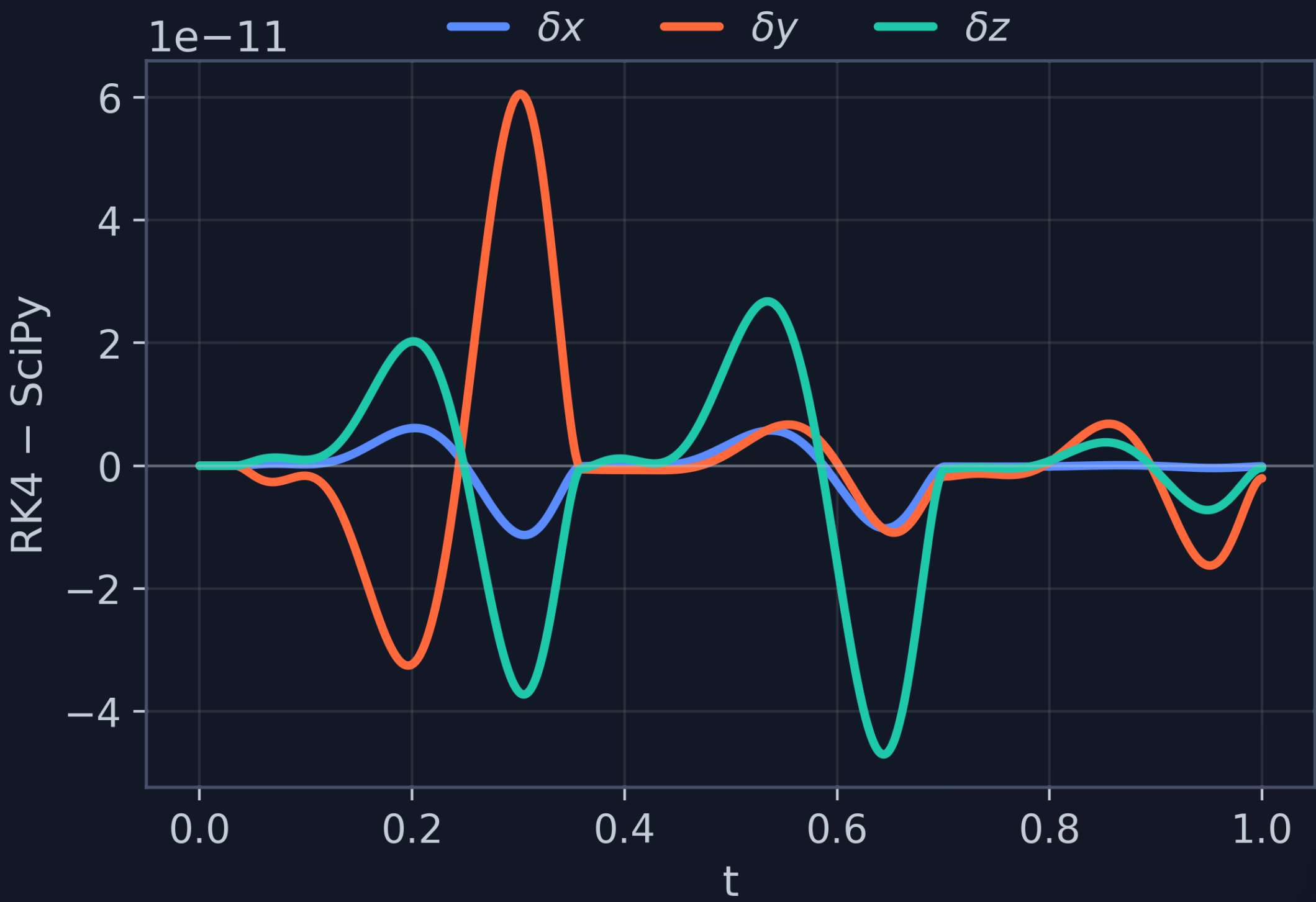


Select
best

Numerical Baseline & Validation

Result

Two independent solvers build the reference and cross-check it **before any ANN sees it**: a custom **RK4** integrator ($h = 10^{-8}$, 1001 points) and **SciPy DOP853** ($rtol = 10^{-10}$), split 80/20 for train–validation (seed 42).



Pointwise RK4 - SciPy disagreement stays at the 10^{-11} level across the trajectory.

1.8×10^{-22}

solver-agreement MSE (RK4 vs SciPy)

✓ target was $< 10^{-30}$

The **ground truth** for every ANN we train in FYDP-II.

Status (FYDP-I) & Future Work

Delivered this stage

- Research question framed; scope kept narrow
- 17-paper review with an explicit gap
- Two-phase design (69 runs) locked
- Dual-solver baseline verified

Next: FYDP-II

- Build the training pipeline; run Phase 1 (60), then Phase 2 (9)
- Repeat seeds for stability; denser grids & new initial conditions
- Compare directly with PINN / operator-learning baselines

We make **no accuracy claims** until the FYDP-II experiments produce evidence. The baseline and the design are locked; the measurements come next.

69

controlled training runs across both phases

1001

reference points from the RK4 integrator

17

papers reviewed to locate the gap

5

evaluation metrics for accuracy & cost

Selected References

- [1] Lorenz (1960), *Tellus* 12(3).
[2] Matthews & Bihlo (2024), PinNDE.
[3] Raissi et al. (2019), *JCP* 378.

- [4] Lagaris et al. (1998), *IEEE TNN* 9(5).
[5] Lau & Werth (2020), ODEN.
[6] Chen et al. (2018), *Neural ODEs, NeurIPS*.

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github.com/ihmorad1/fydp_workspace